URDF Model Quick Start

DR-xcobo 2.0 | v1.0 | ROS 1 / RViz

Load the robot model in RViz and explore its joints using interactive sliders. Keep the complete ZIP: the URDF needs all 10 STL mesh files to display the robot.

Environment

Ubuntu 20.04 LTS Desktop (Focal) + ROS 1 Noetic Ninjemys
Tools: RViz, catkin, and a Bash terminal.

Start with ROS Noetic and its package repository already installed and configured. Use the instructor-provided environment if available. This is a ROS 1 package; the commands below do not apply directly to ROS 2.

Target environment, not yet runtime-tested. The package files and mesh references have been checked. Noetic reached end of life on May 31, 2025; this guide documents the environment for the existing teaching package. Official sources are linked on page 2.

1. Install the required tools

In an Ubuntu Bash terminal, load ROS and install the visualization dependencies:

```
source /opt/ros/noetic/setup.bash
sudo apt update
sudo apt install build-essential unzip ros-noetic-catkin \
  ros-noetic-robot-state-publisher \
  ros-noetic-joint-state-publisher \
  ros-noetic-joint-state-publisher-gui ros-noetic-rviz
```

If /opt/ros/noetic/setup.bash is missing, obtain or install the course ROS environment first. These commands do not install a complete ROS distribution.

2. Download and build

Download the URDF ZIP (3.8 MB) and save it in **~/Downloads/**. Extract the complete package into the workspace:

```
mkdir -p ~/catkin_ws/src
unzip ~/Downloads/dr-xcobo-2.0-urdf-v1.0.zip -d ~/catkin_ws/src
cd ~/catkin_ws
catkin_make
source devel/setup.bash
```

The package path should be **~/catkin_ws/src/Dr_arm_6_desk/**, with **package.xml** directly inside it. Keep the package name and meshes folder unchanged. If using another catkin workspace, substitute its path. Extract the ZIP only once.

Launch & Explore

DR-xcobo 2.0 | RViz visualization

3. Start RViz and the joint sliders

In a new Bash terminal, load both environments, then launch the package:

```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash
roslaunch Dr_arm_6_desk display.launch
```

Two windows should open: **RViz** and **Joint State Publisher GUI**. The launch file also starts Robot State Publisher and starts the ROS master if needed. RViz is configured with **base_link** as the fixed frame.

For a static pose without the slider window, use:

```
roslaunch Dr_arm_6_desk display.launch gui:=false
```

4. Inspect the model

Camera: use RViz's Move Camera tool to orbit, pan, and zoom.
Joints: drag the sliders in the separate GUI to inspect direction and range.
Frames: select **Add > TF** in RViz to show the link coordinate frames.
Stop: press **Ctrl+C** in the terminal that launched the package.

The model has 10 links and 9 joints: 6 revolute, 2 prismatic, and 1 fixed. The two finger joints are independently controlled. End-effector planning requires MoveIt; gravity and contact simulation require a separate simulator setup.

Troubleshooting

Package not found: check the folder path, run catkin_make, and source ~/catkin_ws/devel/setup.bash in the current terminal.

Robot or meshes missing: verify that the full meshes folder was extracted. Enable RobotModel and set Robot Description to robot_description.

Fixed frame error: set Global Options > Fixed Frame to base_link and check the launch terminal for node errors.

Slider window missing: install ros-noetic-joint-state-publisher-gui and launch without gui:=false.

Model scope: the original URDF and meshes are unchanged. Movable joints have zero effort and velocity limits; review these before planning, controller use, or physical simulation. This download provides model visualization only.

Resources

Course resources | Complete URDF ZIP
ROS REP 3: Noetic target platforms | Official Noetic end-of-life notice